

Definition 1.1 We define a Laurent series expanded from the point $z_0 \in \mathbb{C}$ a series of the form

$$\sum_{n=-\infty}^{+\infty} a_n (z - z_0)^n := \sum_{n=0}^{+\infty} a_n (z - z_0)^n + \sum_{n=1}^{+\infty} a_{-n} (z - z_0)^{-n}.$$

The domain of convergence of the Laurent series is defined as the intersection of the domains of convergence of the series on the right hand side of the equation above.

Theorem 1.1 LAURENT'S THEOREM. Assume that $f(z)$ is analytic in an open annulus

$$\{z \in \mathbb{C} \mid r < |z - z_0| < R\}.$$

Then the corresponding Laurent series in this annulus is uniquely determined by

$$f(z) = \sum_{n=-\infty}^{+\infty} a_n (z - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz, \quad \text{for every } n \in \mathbb{Z},$$

and where C is any simple, closed curve separating $|z - z_0| = r$ from $|z - z_0| = R$, run through in the positive sense of the plane.

$$f(z) = \sum_{n=-\infty}^{+\infty} a_n z^n, \quad \text{for } |z| > R.$$

Then we have the same three possibilities as above for a finite isolated singularity, though it here are the *positive* exponents which are causing troubles:

- 1) If $a_n = 0$ for every $n \in \mathbb{N}$, then ∞ is a *removable singularity* for $f(z)$. In this case we define by continuous expansion,

$$f(\infty) = a_0.$$

We note that we in connection with the $\mathfrak{z}$ -transform always consider Laurent series of this type.

- 2) If $a_n \neq 0$ for some, though only finitely many $n \in \mathbb{N}$, e.g. $a_N \neq 0$ and $a_n = 0$ for every $n > N$, then we call ∞ a *pole* of $f(z)$ of order N .
- 3) Finally, if $a_n \neq 0$ for infinitely many $n \in \mathbb{N}$, we call ∞ an *essential singularity* of $f(z)$.

We should here add that e.g. $\sin z \neq 0$ for $z = n\pi$, $n \in \mathbb{Z}$, then $z = n\pi$, $n \in \mathbb{Z}$, are poles of $1/\sin z$. However, since $z = n\pi \rightarrow \infty$ for $n \rightarrow \pm\infty$, we see that ∞ is *not* an isolated singularity of $1/\sin z$, and it is not possible later on to speak about the residue at ∞ for such functions.

Example 2.1 Find the Laurent series expansions of the function

$$f(z) = \frac{1}{z-2}, \quad z \neq 2,$$

from $z_0 = 0$ in each of the domains in which there exists such an expansion.

1) In the disc $|z| < 2$ the constant 2 is dominating in the denominator, and $\left|\frac{z}{2}\right| < 1$. Hence,

$$f(z) = \frac{1}{z-2} = -\frac{1}{2} \cdot \frac{1}{1-\frac{z}{2}} = -\frac{1}{2} \sum_{n=0}^{+\infty} \left\{\frac{z}{2}\right\}^n = -\sum_{n=0}^{+\infty} \frac{1}{2^{n+1}} z^n, \quad \text{for } |z| < 2,$$

which clearly is a Taylor series.

2) We have in the complementary of a disc, $|z| > 2$, that $\left|\frac{1}{z}\right| < 1$, so in this case we instead use that

$$f(z) = \frac{1}{z-2} = \frac{1}{z} \cdot \frac{1}{1-\frac{2}{z}} = \frac{1}{z} \sum_{n=0}^{+\infty} \left\{\frac{2}{z}\right\}^n = \sum_{n=1}^{+\infty} 2^{n-1} \cdot \frac{1}{z^n}, \quad \text{for } |z| > 2.$$

Example 2.2 Find the Laurent series expansions of the function

$$f(z) = \frac{1}{(z-1)(z-2)}, \quad z \in \mathbb{C} \setminus \{1, 2\},$$

from $z_0 = 0$ in each of the domains where such an expansion exists.

$$f(z) = \frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1},$$

1) If $z \in \Omega_1 = \{z \in \mathbb{C} \mid |z| < 1\}$, then we get the Taylor series

$$f(z) = \frac{1}{z-2} - \frac{1}{z-1} = -\frac{1}{2} \cdot \frac{1}{1-\frac{z}{2}} + \frac{1}{1-z} = \sum_{n=0}^{+\infty} \left\{1 - \frac{1}{2^{n+1}}\right\} z^n, \quad \text{for } |z| < 1.$$

We note that there are no negative exponents in this expansion.

2) If $z \in \Omega_2 = \{z \in \mathbb{C} \mid 1 < |z| < 2\}$, then we get the Laurent series

$$f(z) = \frac{1}{z-2} - \frac{1}{z-1} = -\frac{1}{2} \cdot \frac{1}{1-\frac{1}{z}2} - \frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\sum_{n=0}^{+\infty} \frac{1}{2^{n+1}} z^n - \sum_{n=1}^{+\infty} \frac{1}{z^n}.$$

We note that we have both positive and negative exponents in the Laurent series expansion.

Example 2.11 Find the Laurent series for

$$\frac{1}{z^2(1-z)}$$

in the sets $0 < |z| < 1$, and $|z| > 1$.

(a) If $0 < |z| < 1$, then

$$\frac{1}{z^2(1-z)} = \frac{1}{z^2} \sum_{n=0}^{+\infty} z^n = \sum_{n=-2}^{+\infty} z^n.$$

(b) If $|z| > 1$, then

$$\frac{1}{z^2(1-z)} = -\frac{1}{z^3} \cdot \frac{1}{1-\frac{1}{z}} = -\frac{1}{z^3} \sum_{n=0}^{+\infty} \frac{1}{z^n} = -\sum_{n=3}^{+\infty} \frac{1}{z^n}.$$

Example 2.12 Find the Laurent series expansion from 0 for each of the following functions in the given domains:

$$(a) \frac{z^2-1}{(z+2)(z+3)} \text{ for } |z| > 3, \quad (b) \frac{24}{z^2(z-1)(z+2)} \text{ for } 0 < |z| < 1.$$

(a) Since $\left|\frac{2}{z}\right| < 1$ and $\left|\frac{3}{z}\right| < 1$ for $|z| > 3$, we get by a decomposition (remember the constant term, because the numerator and the denominator have the same degree),

$$\begin{aligned} \frac{z^2-1}{(z+2)(z+3)} &= 1 + \frac{3}{z+2} - \frac{8}{z+3} = 1 + \frac{3}{z} \cdot \frac{1}{1+\frac{2}{z}} - \frac{8}{z} \cdot \frac{1}{1+\frac{3}{z}} \\ &= 1 + \frac{3}{z} \sum_{n=0}^{+\infty} (-1)^n \cdot \frac{2^n}{z^n} - \frac{8}{z} \sum_{n=0}^{+\infty} (-1)^n \cdot \frac{3^n}{z^n} \\ &= 1 + \sum_{n=1}^{+\infty} (-1)^{n-1} \cdot \frac{3 \cdot 2^{n-1}}{z^n} - \sum_{n=1}^{+\infty} (-1)^{n-1} \cdot \frac{8 \cdot 3^{n-1}}{z^n} \\ &= 1 + \sum_{n=1}^{+\infty} (-1)^n \{8 \cdot 3^{n-1} - 3 \cdot 2^{n-1}\} \cdot \frac{1}{z^n}. \end{aligned}$$

(b) Since $|z| < 1$ and $\left|\frac{z}{2}\right| < 1$ for $0 < |z| < 1$, it follows by a decomposition, in which $\frac{1}{z^2}$ does NOT

enter that

$$\begin{aligned}
 \frac{24}{z^2(z-1)(z+2)} &= \frac{1}{z^2} \left\{ \frac{8}{z-1} - \frac{8}{z+2} \right\} = -\frac{1}{z^2} \left\{ \frac{8}{1-z} + \frac{4}{1+\frac{z}{2}} \right\} \\
 &= -\frac{1}{z^2} \left\{ \sum_{n=0}^{+\infty} 8z^n + \sum_{n=0}^{+\infty} (-1)^n \cdot 4 \cdot \frac{1}{2^n} z^n \right\} = -\sum_{n=0}^{+\infty} \left\{ 8 + (-1)^n \frac{4}{2^n} \right\} z^{n-2} \\
 &= -\sum_{n=-2}^{+\infty} \left\{ 8 + (-1)^n \cdot \frac{1}{2^n} \right\} z^n.
 \end{aligned}$$

Example 2.29 Given the function

$$f(z) = \frac{1}{z^2(z-4)}.$$

1) Find the Laurent series of f in the annulus $0 < |z| < 4$.

2) Find the Laurent series of f in the set $4 < |z|$.

3) Compute the integrals

$$\oint_{|z|=1} f(z) dz \quad \text{and} \quad \oint_{|z|=5} f(z) dz.$$

1) If $0 < |z| < 4$, then $\left| \frac{z}{4} \right| < 1$, hence

$$\begin{aligned}
 f(z) &= \frac{1}{z^2(z-4)} = -\frac{1}{4z^2} \cdot \frac{1}{1-\frac{z}{4}} = -\frac{1}{4z^2} \sum_{n=0}^{+\infty} \frac{z^n}{4^n} = \sum_{n=0}^{+\infty} \left(-\frac{1}{4^{n+1}} \right) z^{n-2} \\
 &= \sum_{n=-2}^{+\infty} \left(-\frac{1}{4^{n+3}} \right) z^n, \quad 0 < |z| < 4.
 \end{aligned}$$

2) If instead $|z| > 4$, then $\left| \frac{4}{z} \right| < 1$, hence

$$f(z) = \frac{1}{z^2(z-4)} = \frac{1}{z^3} \cdot \frac{1}{1-\frac{4}{z}} = \frac{1}{z^3} \sum_{n=0}^{+\infty} \frac{4^n}{z^n} = \sum_{n=3}^{+\infty} \frac{4^{n-3}}{z^n}.$$

3) The closed curve $|z| = 1$ lies in the annulus $0 < |z| < 4$, so if we apply the Laurent series from (1), we get

$$\oint_{|z|=1} f(z) dz = 2\pi i a_{-1} = 2\pi i \cdot \left(-\frac{1}{4^2} \right) = -\frac{\pi i}{8}.$$

The closed curve $|z| = 5$ lies in the set $|z| > 4$, so if we apply the Laurent series from (2), we get

$$\oint_{|z|=5} f(z) dz = 2\pi i \tilde{a}_{-1} = 0.$$